

# EE3621: Digital Signal Processing

## Lecture 6: Inverse Z-Transform & Stability Analysis (III B.Tech EEE)

Lecture Notes & Comprehensive Worked Numerical

### 1 Lecture Plan & Objectives (40 Minutes Breakdown)

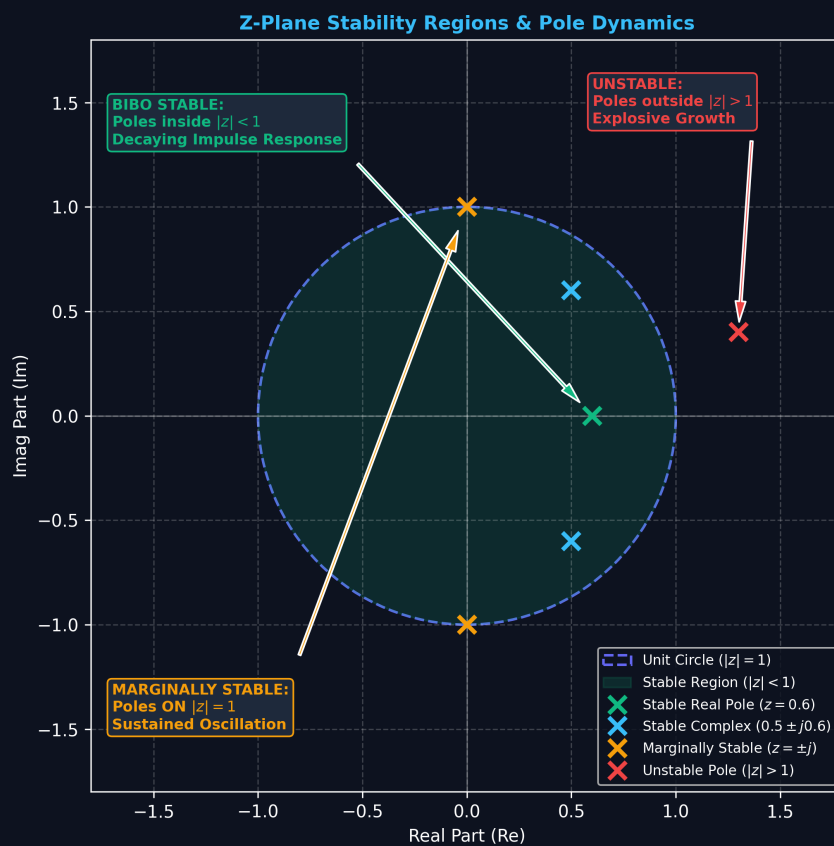


Figure 1: Z-Plane Stability Regions & Pole Dynamics: Stable ( $|z| < 1$ ), Marginally Stable ( $|z| = 1$ ), Unstable ( $|z| > 1$ ).

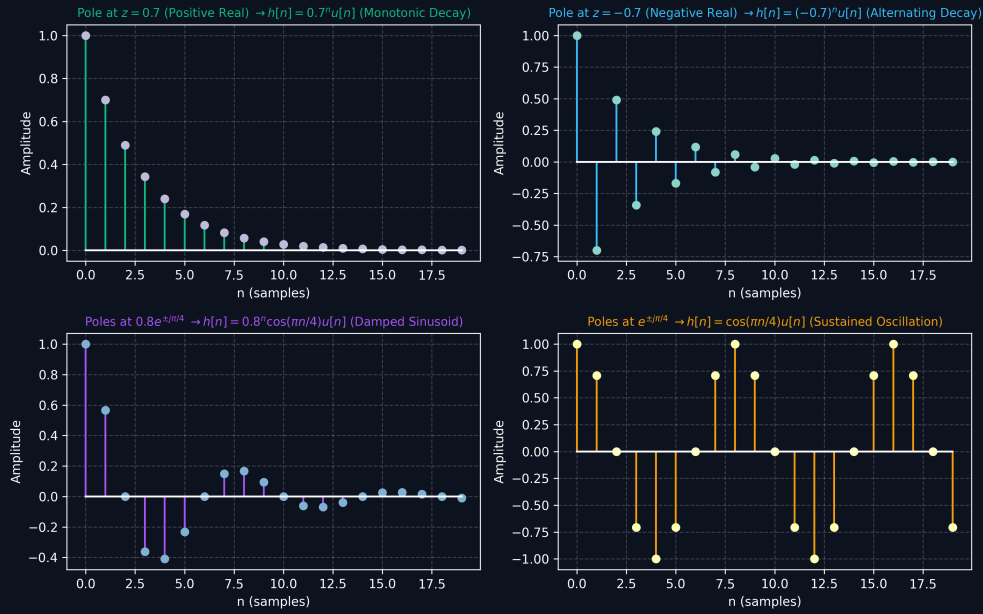


Figure 2: Time-Domain Impulse Response Modes corresponding to Pole Locations in the Complex Z-Plane.

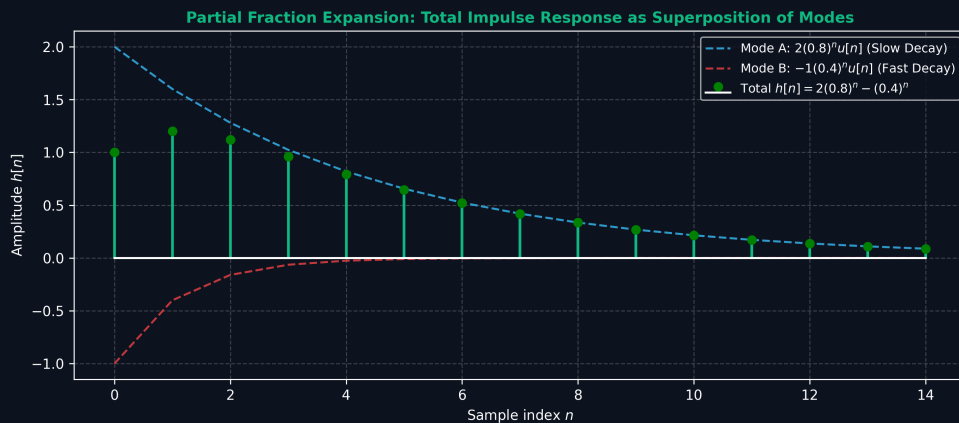


Figure 3: Partial Fraction Expansion: Synthesizing the Complete Impulse Response as a Superposition of Modes.

- **00:00 – 05:00 (5 mins):** Motivation: Why do we need the Inverse Z-transform? Uniqueness of ROC.
- **05:00 – 15:00 (10 mins):** Method 1: Partial Fraction Expansion (Distinct and Repeated Poles).
- **15:00 – 22:00 (7 mins):** Method 2: Power Series Expansion (Long Division for causal vs. anti-causal).
- **22:00 – 28:00 (6 mins):** Method 3: Contour Integration (Residue Theorem).
- **28:00 – 36:00 (8 mins):** Comprehensive Worked Numerical comparing all 3 approaches.
- **36:00 – 40:00 (4 mins):** Stability & Causality Analysis of LTI Systems based on poles.

## 2 Overview of the Inverse Z-Transform

The Inverse Z-transform maps  $X(z)$  and its ROC back to a discrete sequence  $x[n]$ :

$$x[n] = \mathcal{Z}^{-1}\{X(z)\} \quad (1)$$

Specification of the ROC is critical to ensure uniqueness.

## 3 Method 1: Partial Fraction Expansion (PFE)

For distinct poles  $p_k$ :

$$\frac{X(z)}{z} = \sum_{k=1}^N \frac{A_k}{z - p_k} \quad \text{where} \quad A_k = \left[ (z - p_k) \frac{X(z)}{z} \right]_{z=p_k} \quad (2)$$

## 4 Method 2: Power Series Expansion (Long Division)

We express  $X(z)$  directly as a Laurent series:

$$X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n} = \dots + x[-1]z^1 + x[0] + x[1]z^{-1} + x[2]z^{-2} + \dots \quad (3)$$

- **Causal** ( $|z| > |p|$ ): Divide in descending powers of  $z$  ( $x[0] + x[1]z^{-1} + x[2]z^{-2} + \dots$ ).
- **Anti-causal** ( $|z| < |p|$ ): Divide in ascending powers of  $z$  ( $x[-1]z^1 + x[-2]z^2 + \dots$ ).

## 5 Method 3: Contour Integration (Cauchy's Residue Theorem)

$$x[n] = \frac{1}{2\pi j} \oint_C X(z)z^{n-1} dz = \sum \text{Residues of } [X(z)z^{n-1}] \text{ inside } C \quad (4)$$

For a simple pole at  $z = p$ :

$$\text{Res} \{ X(z)z^{n-1} \}_{z=p} = \lim_{z \rightarrow p} (z - p)X(z)z^{n-1} \quad (5)$$

## 6 Comprehensive Worked Numerical: All 3 Approaches

### 6.1 Problem Statement

Given the rational Z-transform:

$$X(z) = \frac{1}{1 - 1.5z^{-1} + 0.5z^{-2}} = \frac{z^2}{(z - 1)(z - 0.5)} \quad (6)$$

Determine the sequence  $x[n]$  for:

1. **Condition 1:** Causal Sequence (ROC:  $|z| > 1$ )
2. **Condition 2:** Anti-Causal Sequence (ROC:  $|z| < 0.5$ )
3. **Condition 3:** Two-Sided / BIBO Stable Sequence (ROC:  $0.5 < |z| < 1$ )

## 6.2 Approach 1: Partial Fraction Expansion

Form  $\frac{X(z)}{z} = \frac{z}{(z-1)(z-0.5)} = \frac{A}{z-1} + \frac{B}{z-0.5}$ . Residues:

$$A = \frac{z}{z-0.5} \Big|_{z=1} = \frac{1}{1-0.5} = 2, \quad B = \frac{z}{z-1} \Big|_{z=0.5} = \frac{0.5}{0.5-1} = -1 \quad (7)$$

Reconstructing  $X(z)$ :

$$X(z) = 2 \left( \frac{z}{z-1} \right) - 1 \left( \frac{z}{z-0.5} \right) = \frac{2}{1-z^{-1}} - \frac{1}{1-0.5z^{-1}} \quad (8)$$

Applying ROC conditions:

- **Causal** ( $|z| > 1$ ):  $x[n] = 2(1)^n u[n] - (0.5)^n u[n] = [2 - (0.5)^n] u[n]$ .  
Values:  $x[0] = 1.0$ ,  $x[1] = 1.5$ ,  $x[2] = 1.75$ ,  $x[3] = 1.875$ .
- **Anti-Causal** ( $|z| < 0.5$ ):  $x[n] = -2(1)^n u[-n-1] + (0.5)^n u[-n-1] = [(0.5)^n - 2] u[-n-1]$ .  
Values:  $x[-1] = 0$ ,  $x[-2] = 2$ ,  $x[-3] = 6$ ,  $x[-4] = 14$ .
- **Two-Sided** ( $0.5 < |z| < 1$ ):  $x[n] = -(0.5)^n u[n] - 2u[-n-1]$ .

## 6.3 Approach 2: Power Series Expansion (Long Division)

**Case A: Causal Sequence** ( $|z| > 1$ )

Divide 1 by  $(1 - 1.5z^{-1} + 0.5z^{-2})$ :

- Step 1:  $1/1 = \mathbf{1}$ . Remainder:  $1.5z^{-1} - 0.5z^{-2}$ .
- Step 2:  $1.5z^{-1}/1 = \mathbf{1.5z^{-1}}$ . Remainder:  $1.75z^{-2} - 0.75z^{-3}$ .
- Step 3:  $1.75z^{-2}/1 = \mathbf{1.75z^{-2}}$ . Remainder:  $1.875z^{-3} - 0.875z^{-4}$ .

Quotient:  $X(z) = 1 + 1.5z^{-1} + 1.75z^{-2} + 1.875z^{-3} + \dots \implies x[0] = 1, x[1] = 1.5, x[2] = 1.75, x[3] = 1.875$ .

**Case B: Anti-Causal Sequence** ( $|z| < 0.5$ )

Divide 1 by  $(0.5z^{-2} - 1.5z^{-1} + 1)$ :

- Step 1:  $1/0.5z^{-2} = \mathbf{2z^2}$ . Remainder:  $3z - 2z^2$ .
- Step 2:  $3z/0.5z^{-2} = \mathbf{6z^3}$ . Remainder:  $7z^2 - 6z^3$ .
- Step 3:  $7z^2/0.5z^{-2} = \mathbf{14z^4}$ .

Quotient:  $X(z) = 0z^1 + 2z^2 + 6z^3 + 14z^4 + \dots \implies x[-1] = 0, x[-2] = 2, x[-3] = 6, x[-4] = 14$ .

## 6.4 Approach 3: Contour Integration (Residue Theorem)

Integrand  $F(z) = X(z)z^{n-1} = \frac{z^{n+1}}{(z-1)(z-0.5)}$ .

Residues at simple poles:

$$\text{Res}\{F(z)\}_{z=1} = \frac{z^{n+1}}{z-0.5} \Big|_{z=1} = 2(1)^n, \quad \text{Res}\{F(z)\}_{z=0.5} = \frac{z^{n+1}}{z-1} \Big|_{z=0.5} = -(0.5)^n \quad (9)$$

- **Causal** ( $|z| > 1$ ): Encloses  $z = 1, 0.5 \implies x[n] = \text{Res}_{z=1} + \text{Res}_{z=0.5} = [2 - (0.5)^n] u[n]$ .
- **Anti-Causal** ( $|z| < 0.5$ ): Evaluates exterior residues  $\implies x[n] = -(\text{Res}_{z=1} + \text{Res}_{z=0.5}) = [(0.5)^n - 2] u[-n-1]$ .
- **Two-Sided** ( $0.5 < |z| < 1$ ): Encloses only  $z = 0.5 \implies x[n] = -(0.5)^n u[n] - 2u[-n-1]$ .

## 7 Causality, Stability & Pole Locations

- **Causality:** The ROC of a causal LTI system transfer function  $H(z)$  must be the exterior of a circle:  $|z| > p_{max}$ .
- **Stability:** The ROC of  $H(z)$  must contain the unit circle ( $|z| = 1$ ).
- **Causal & Stable:** All poles must lie strictly inside the unit circle ( $|p_k| < 1$ ).